CS_5080_Lecture_9

Tuesday, February 24, 2026 4:43 PM

* Chap 5.5. continuation *

ρ = row

(actual)

off-policy RL algor

behavior policy b: (acts)

Transfer

target policy π:

eps 1

o - - - (x) - - - (o)

$G_t | s$

$G_t | s, a$

ρ

eps i - - - - - eps n

o - - - (x) - - - o - - - (o)

$G_t | x$

$G_t | x, a$

ρ

$G_t | x$

$G_t | x, a$

ρ

Transfers / we need "important sampaling"

o - - - (x) - - - (o)

(imagination)

τ = TAU

ordinary important sampling

$$V_\pi(s) = \frac{\sum_{t \in \tau(s)} \rho_t G_t}{|\tau(s)|}$$

where $\tau(s)$ = set of time points in which state s occurs in behavior b

G_t = return at state s (if learning $V_\pi(s)$)

or = return for (s,a) pair (" " $q_\pi(s,a)$)

we know

$$\rho_t = \prod_{i=1} \frac{\pi(a_k | s_k)}{b(a_k | s_k)}$$

weighted important sampling

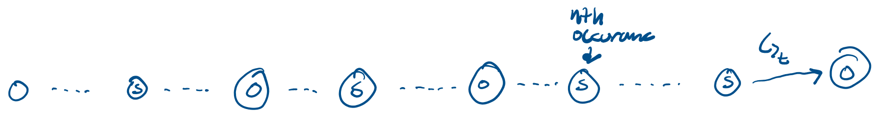
$$V_\pi(s) = \frac{\sum_{t \in \tau(s)} \rho_t G_t}{\sum_{t \in \tau(s)} \rho_t}$$

* looking at 5.6 Incremental implementation *

we have computed current $V_\pi(s)$ up to the $(n-1)$th occurance of state s in behavior b. We find state s the nth time. How to obtain

$V_{n+1}(s)$ given $V_{n,\pi}(s)$

and G_t of the last occurance



π : $V_n(s) = \dfrac{\sum_{k=1}^{n-1} \rho_t G_t}{\sum_{k=1}^{n-1} \rho_t}$ ---- ①

$V_{n+1}(s) = \dfrac{\sum_{k=1}^{n} \rho_t G_t}{\sum_{k=1}^{n} \rho_t}$ ---- ②

compute

$$V_{n+1}(s) - V_n(s) = \frac{\sum_{i=1}^{n} \rho_t G_t}{\sum_{i=1}^{n} \rho_t} - \frac{\sum_{i=1}^{n-1} \rho_t G_t}{\sum_{i=1}^{n-1} \rho_t}$$

$$\frac{\left(\sum_{i=1}^{n} \rho_t G_t\right)\left(\sum_{i=1}^{n-1} \rho_t\right) - \left(\sum_{i=1}^{n-1} \rho_t G_t\right)\left(\sum_{i=1}^{n} \rho_t\right)}{\left(\sum_{i=1}^{n} \rho_t\right)\left(\sum_{i=1}^{n-1} \rho_t\right)}$$

$$= \frac{\left(\sum_{i=1}^{n-1} \rho_t G_t + \rho_n G_n\right)\left(\sum_{i=1}^{n-1} \rho_t\right) - \left(\sum_{i=1}^{n-1} \rho_t G_t\right)\left(\sum_{i=1}^{n-1} \rho_t + \rho_n\right)}{\left(\sum_{i=1}^{n} \rho_t\right)\left(\sum_{i=1}^{n-1} \rho_t\right)}$$

$$= \frac{\rho_n G_n \sum_{i=1}^{n-1} \rho_t - \rho_n \sum_{i=1}^{n-1} \rho_t G_t}{(\;''\;)\;(\;''\;)}$$

$$V_{n+1}(s) - V_n(s) = \frac{\rho_n}{\sum^{n} \rho_i}\left[G_n - V_n(s)\right]$$

$$V_{n+1}(s) - V_n(s) = \frac{\rho_n}{\sum_{i=1}^{n} \rho_i} \left[G_n - V_n(s) \right]$$

$$V_{n+1}(s) = V_n(s) + \frac{\rho_n}{\sum_{i=1}^{n} \rho_i} \left[G_n - V_n(s) \right]$$